AKASH R JADHAV

Software Developer

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SUMMARY

software developer (B.Tech CSE, PES University, CGPA 8.00) with internship experience in system design, software architecture, debugging and test case design. Built a Java shared-memory telemetry prototype backed by 30+ JUnit tests, a network security pipeline with pytest and Jenkins regression testing, and a verifier that checks LLM-generated text against source evidence.

SKILLS

Languages: Java, Python, C++, JavaScript

Software Design & Systems: System design, software architecture, data structures & algorithms, shared-memory/IPC design, lock-free ring buffers, TCP/IP packet-level networking (Scapy), OpenTelemetry

Testing & Debugging: Debugging multi-JVM and multi-container systems, code review (Git), test case design, JUnit, pytest, Maven, Jenkins CI/CD

AI & LLM Systems: RAG, FAISS, vector embeddings, hybrid retrieval, FastMCP, LLM output verification, PyTorch, TensorFlow/Keras

Back-End & Databases: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, MySQL, MongoDB, SQLite

Cloud & DevOps: Docker, AWS Lambda, EventBridge, Prometheus, Loki, Grafana

EXPERIENCE

Developer Intern, CitiusTech (Client: Evident Microscopy) — Bangalore **May 2026 – Aug 2026**

- Contributed to a prototype (not production) Java/OpenTelemetry observability pipeline for a client sample log of 120+ system components (~60,000 log lines/hour), feeding Prometheus, Loki and 60+ custom Grafana dashboards
- Designed the shared-memory ingestion path: multiple writers into an Agrona ManyToOneRingBuffer over a memory-mapped file, read by a separate JVM; ~3 ms writer-to-ring-buffer ingest time
- Justified shared memory over Kafka and Chronicle Queue to isolate applications from collector slowness; added OpenTelemetry Java agent auto-instrumentation via JVM options, with no component code changes
- Wrote 30+ JUnit tests (integration and reader/writer synchronization, no data loss) run with Maven; debugged a writer/reader rate mismatch across JVMs; used Git for version control and code review

Research Intern, PESU C-IoT Research Lab — Bangalore **Jan 2026 – Mar 2026**

- Built an automated ingestion and feature-engineering pipeline on ICAR-CRIDA CropPest DSS data (16 features, 1,886 weekly records, 6 agro-climatic locations) to forecast next-week pest outbreak risk
- Designed a privacy-preserving federated learning architecture (PyTorch MLP, Flower, Ray) using FedProx for Non-IID client drift and FedPer with personalized per-client classification heads, without pooling raw data
- Evaluated the models against baselines: FedPer reached 81.05% peak test accuracy vs. 77.54% for FedProx; a multi-pest ensemble reached 70.76%

PROJECTS

AI Job Agent v2 *(Python, FastAPI, FAISS, Gemini embeddings, FastMCP, AWS Lambda, pytest)*

- Built serverless ingestion (AWS Lambda + EventBridge, 8-hour cycle, Docker) across 7 job sources with title normalization, deduplication and description hashing, persisted via async SQLAlchemy to SQLite
- Designed a RAG layer over 40+ evidence chunks: gemini-embedding-001 (768-d), FAISS IndexFlatIP, and hybrid retrieval fusing dense similarity with keyword overlap via Reciprocal Rank Fusion (k=60); exposed through a FastMCP server with 7 tools
- Built a dual-layer verifier for LLM-generated text (deterministic tech/language/metric checks plus a verb-inflation check against source evidence), covered by 36+ hermetic pytest tests

Network Security Validation Pipeline *(Python, Scapy, Docker, pytest, Jenkins, SQLite)*

- Built attack simulation and detection modules for ARP Spoofing, MAC Flooding and Rogue DHCP using raw TCP/IP packet crafting (Scapy) across an isolated 3-container Docker bridge network
- Designed a multi-layer heuristic detection engine combining 4 independent signal layers with weighted scoring, alerting only on multi-signal convergence
- Automated regression testing with pytest, logged results to SQLite, and integrated a Jenkins CI/CD pipeline via GitHub webhooks for validation on every push

Hybrid Federated and Centralized Deepfake Detection System *(Python, TensorFlow/Keras, PyTorch, FastAPI, Scikit-learn)*

- Built a privacy-preserving deepfake detection web app achieving 82% accuracy using XceptionNet and EfficientNet, with ONNX for cross-platform model interoperability and local inference
- Implemented Gemini API–based misinformation verification for detected synthetic media

SMARTBUS — Reservation System *(Python, Streamlit, MySQL)*

- Engineered a Streamlit-based booking platform on a 12+ table normalized MySQL schema, using triggers and CASCADE constraints for automated seat allocation and data integrity
- Optimized concurrent booking transactions to prevent seat conflicts

EDUCATION

B.Tech in Computer Science and Engineering, PES University (EC Campus) **Graduated 2026**

CGPA: 8.00 | Class XII: 92.7% (KCET Rank 5759) | Class X: 87.4%

CERTIFICATIONS, TRAINING & AWARDS

- Oracle Cloud Infrastructure 2025 Multicloud Architect Professional and Foundations Associate — Oracle University
- AWS Educate: Getting Started with Compute (Trained) — Amazon Web Services
- AWS Educate: Getting Started with Serverless (Trained) — Amazon Web Services
- Academic Distinction Award — 7 Semesters